

SAVVYDFIR-MCP Investigation Report

ROCBA-2020-FREDS-LAPTOP · Generated 2026-06-08T01:14:01.857478+00:00

Summary

Investigation **ROCBA-2020-FREDS-LAPTOP** reached triage status **COMPLETE_WITH_GAPS** with **2** structurally confirmed finding(s), 2 of which are corroborated by multiple independent artifact sources, and **125** active lead(s) pending validation.

Confirmed activity (full detail in Top Confirmed Findings below):

- Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry, confidence 0.90) confirms the attacker's Outlook PST email exfiltration to Google Drive...
- Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch, confidence 0.95) confirms the attacker's active SRL IP data collection and physical...

Hunt closed: 2 confirmed (pre_planned_physical_intrusion_staged_account; multi_channel_exfiltration_usb_cloud_lateral_movement); 3 suspended (bitlocker_recovery_key_exfiltration; icloud_opportunistic_cloud_sync_exfiltration; anti_forensics_evidence_destruction_multi_method).

Cross-artifact synthesis: the confirmed activity is corroborated across multiple independent artifact sources.

Kill-chain coverage: 5 of 6 phases evidenced, 1 blindspot(s).

Caveats: 5 unresolved cross-artifact discrepancy(ies).

CASE STATUS

COMPLETE

TRIAGE

COMPLETE_WITH 127

FINDINGS

CONFIRMED

2

ACTIVE LEADS

125

CORRECTION EVENTS

5

UNRESOLVED

5

SIGMA HITS

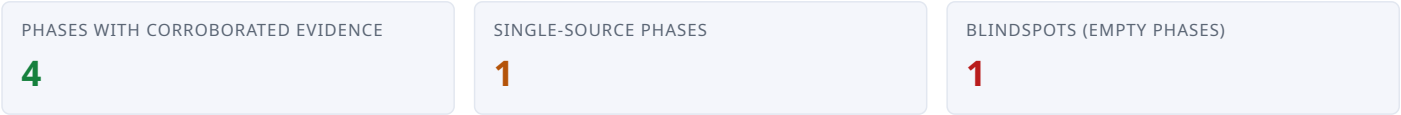
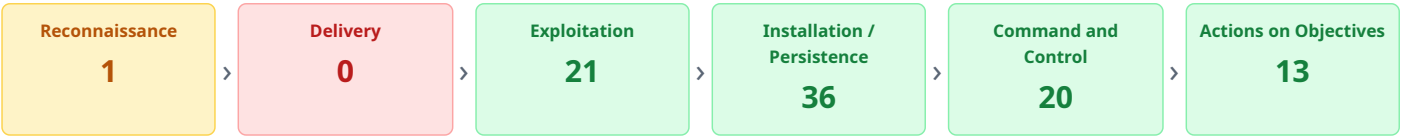
1

ATT&CK COVERAGE

35.7%

Activity Thread - Cyber Kill Chain Coverage

Per Diamond Model Axiom 4, every malicious activity traverses a succession of kill-chain phases. Empty phases below represent **blindspots**: areas where evidence was either not extracted, not retained, or actively destroyed by anti-forensics. Phase color reflects the number of findings mapped to that phase.



KILL-CHAIN PHASE	STATUS
Reconnaissance	1 finding (single source)
Delivery	BLINDSPOT - no evidence
Exploitation	21 findings
Installation / Persistence	36 findings
Command and Control	20 findings
Actions on Objectives	13 findings

ATT&CK Coverage

Tactic-level coverage (covered / gap) is authoritative. Technique and finding columns are inferred from the findings mapped to each tactic.

TACTIC	STATUS	TECHNIQUES OBSERVED	FINDINGS	SUGGESTED NEXT TOOLS
TA0002 Execution	covered	-	-	
TA0003 Persistence	covered	-	-	
TA0005 Defense Evasion	covered	-	-	
TA0008 Lateral Movement	covered	-	-	
TA0011 Command and Control	covered	-	-	
TA0001 Initial Access	gap	-	-	disk.summarize_evtx, memory.scan_network
TA0004 Privilege Escalation	gap	-	-	disk.summarize_evtx, memory.scan_processes
TA0006 Credential Access	gap	-	-	disk.extract_shimcache, disk.get_amcache, disk.summarize_evtx
TA0007 Discovery	gap	-	-	memory.scan_processes, memory.list_dlls, timeline.query_timeline
TA0009 Collection	gap	-	-	disk.extract_prefetch, disk.get_amcache, memory.scan_processes
TA0010 Exfiltration	gap	-	-	disk.extract_srum, memory.scan_network, timeline.query_timeline
TA0040 Impact	gap	-	-	disk.summarize_evtx, memory.detect_injection, timeline.query_timeline
TA0042 Resource Development	gap	-	-	disk.get_amcache, disk.extract_prefetch, timeline.query_timeline
TA0043 Reconnaissance	gap	-	-	disk.summarize_evtx, memory.scan_network, timeline.query_timeline

Key Event Timeline

TIME (UTC)	FINDING	STATUS	EVENT
2020-10-21T03:44:22Z	F-059	ACTIVE	Amcache confirms the srl-h account downloaded SDelete (sdelete64.exe, SHA-1:...
2020-10-21T03:46:19Z	F-118	ACTIVE	srl-h's PowerShell history records sequential execution of sdelete64.exe -nobanner -z -c D: and sdelete64.exe...
2020-10-21T03:46:19Z	F-117	ACTIVE	srl-h PowerShell history (mtime 2020-10-21T03:46:19 UTC) records 6 commands: initial attempts to run sdelete64.exe...
2020-11-13T00:00:01Z	F-047	ACTIVE	NTLM brute force campaign: 58,713 EID 4776 (NTLM validation) and 58,706 EID 4625 (failed logon) events in...
2020-11-14T00:34:00Z	F-122	ACTIVE	Chainsaw Sigma scan (2,284 rules, E-201) against Security.evtx, System.evtx, PowerShell.evtx, and Application.evtx...
2020-11-14T03:42:49Z	F-124	CONFIRMED	Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch,...
2020-11-14T03:42:49Z	F-048	ACTIVE	INBOUND RDP takeover from attacker host 'cobra' (52.249.198.56, Azure IP): At 2020-11-14 03:42:44 UTC (22:42 EST Nov...
2020-11-14T03:43:00Z	F-102	ACTIVE	LNK file analysis confirms the breadth of SRL intellectual property accessed and exfiltrated during the 2020-11-14...
2020-11-14T03:46:00Z	F-101	ACTIVE	CRIMSON2 (F:\) USB drive was used as primary physical exfiltration medium. LNK files created during attack window...
2020-11-14T03:48:53Z	F-106	ACTIVE	Jump Lists confirm the attacker accessed Maria Hill's (mhill) private SharePoint personal documents through Fred's...
2020-11-14T04:13:22Z	F-109	ACTIVE	Edge browser download record at 04:13:22 UTC proves a file from Maria Hill's private SharePoint OneDrive was...
2020-11-14T05:00:44Z	F-049	ACTIVE	Lateral movement from Fred's laptop to shieldbase infrastructure: At 2020-11-14 05:00:44 UTC (00:00 EST) attacker...
2020-11-14T05:05:38Z	F-104	ACTIVE	Jump List for Remote Desktop Connection (AppId: 1bc392b8e104a00e) proves mstsc.exe was used to RDP to...
2020-11-14T12:48:13Z	F-114	ACTIVE	Recycle Bin entry \$IDNBREY.pst proves the BitLocker recovery key file was found on Fred's OneDrive Desktop and...
2020-11-14T13:37:51Z	F-108	ACTIVE	Browser history (Chrome) proves SDelete was searched and downloaded by the attacker during the 2020-11-14 attack...
2020-11-14T13:39:21Z	F-052	ACTIVE	Attacker created backup.pst at 2020-11-14 13:39:21 UTC in Users\fredr\OneDrive\Documents\Outlook Files, wrote email...
2020-11-14T13:42:30Z	F-056	ACTIVE	SDELETE.EXE executed 7 times across two paths during the attacker's session (13:42-13:47 UTC 2020-11-14). First 2...
2020-11-14T14:00:50Z	F-125	CONFIRMED	Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry,...
2020-11-14T14:00:50Z	F-096	ACTIVE	ShellBags confirm Google Drive File Stream was mounted as G:\ and the attacker actively browsed the following G:...
2020-11-14T14:01:34Z	F-105	ACTIVE	Jump Lists confirm Microsoft Outlook 2016 was used to access both the local PST file (backup.pst) and the Google...

TIME (UTC)	FINDING	STATUS	EVENT
2020-11-14T14:01:35Z	F-100	ACTIVE	LNK file SRL-EMAIL-EXPORT.lnk (SourceCreated 2020-11-14 14:01:35 UTC) definitively proves the Outlook PST export was...
2020-11-14T14:04:07Z	F-112	ACTIVE	WordWheelQuery registry key (NTUSER.DAT, LastWriteTimestamp 2020-11-14 14:04:07 UTC) records 11 search terms typed...
2020-11-14T14:07:32Z	F-115	ACTIVE	Recycle Bin contains four post-exfiltration cleanup events establishing the attacker's cleanup sequence: (1)...

Top Confirmed Findings

F-125 CONFIRMED conf 0.97 (high) / T1114.001

Temporal cluster 14:00:50-14:04:07 UTC 2020-11-14 (197s, 3 independent sources: EVTX+file_system+registry, confidence 0.90) confirms the attacker's Outlook PST email exfiltration to Google Drive cloud. Three independent artifact streams co-occur within 4 minutes: EVTX records...

Corroborated by F-050, F-105, F-112, F-100

- file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst
- file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst
- timestamp: 2020-11-14T14:01:34
- timestamp: 2020-11-14T14:03:52
- timestamp: 2020-11-14T14:04:07

F-124 CONFIRMED conf 0.95 (high) / T1005

Temporal cluster 03:42:49-03:46:33 UTC 2020-11-14 (224s, 4 independent sources: EVTX+MFT+file_system+Prefetch, confidence 0.95) confirms the attacker's active SRL IP data collection and physical exfiltration window. Four independent artifact streams co-occur: EVTX records...

Corroborated by F-048, F-043, F-101, F-102

- timestamp: 2020-11-14T03:42:49
- timestamp: 2020-11-14T03:46:33
- executable: CRIMSON2 USB volume mounted
- file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\SRL-Projects

Active Leads & Recommended Pivots

SEVERITY	DETECTOR	CONFIDENCE	DESCRIPTION	RECOMMENDED NEXT PIVOT
MEDIUM	motw_presence	0.70	Mark-of-the-Web Zone.Identifier evidence is present.	Correlate MotW file creation with browser, LNK, Jump List, and execution artifacts.

Active Finding Leads

F-115 ACTIVE conf 0.99 (high) / T1070.004

Recycle Bin contains four post-exfiltration cleanup events establishing the attacker's cleanup sequence: (1) backup.pst (\$IDNBREY.pst) deleted at 2020-11-14 14:07:32 UTC — original path C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst, size 20,587,520 bytes (20.6...

file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst file_path: C:\Users\fredr\Downloads\SDelete\sdelete.exe
timestamp: 2020-11-14T14:07:32Z timestamp: 2020-11-14T13:41:24Z timestamp: 2020-11-14T04:49:53Z

F-114 ACTIVE conf 0.99 (high) / T1552.001

Recycle Bin entry \$IDNBREY.pst proves the BitLocker recovery key file was found on Fred's OneDrive Desktop and deleted to the Recycle Bin at 2020-11-14 12:48:13 UTC. Original path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key...

file_path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key 26F77152-999C-45E8-8BD4-C83FAC7BB72D.TXT
timestamp: 2020-11-14T12:48:13Z

F-109 ACTIVE conf 0.99 (high) / T1213.002

Edge browser download record at 04:13:22 UTC proves a file from Maria Hill's private SharePoint OneDrive was downloaded directly to the CRIMSON2 USB drive. Source URL:...

file_path: F:\Files from SRL system\Quantum Particles Affected by Other Dimensions.pdf
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/Quantum%20Particles%20Affected%20by%20other%20Dimensions.pdf
timestamp: 2020-11-14T04:13:22Z

F-108 ACTIVE conf 0.99 (high) / T1070.004

Browser history (Chrome) proves SDelete was searched and downloaded by the attacker during the 2020-11-14 attack session. At 13:37:37 UTC Chrome shows Google search for "sdelete download"; at 13:37:47 visited docs.microsoft.com/sysinternals/downloads/sdelete; two download...

executable: C:\Users\fredr\Downloads\SDelete.zip file_path: C:\Users\fredr\Downloads\SDelete.zip
timestamp: 2020-11-14T13:37:51Z timestamp: 2020-11-14T13:37:37Z

F-105 ACTIVE conf 0.99 (high) / T1114.001

Jump Lists confirm Microsoft Outlook 2016 was used to access both the local PST file (backup.pst) and the Google Drive PST (SRL-EMAIL-EXPORT.pst). Outlook AppId (Microsoft Outlook 2016 64-bit) entry for backup.pst: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst,...

file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst timestamp: 2020-11-14T14:01:34Z
timestamp: 2020-11-14T14:03:52Z timestamp: 2020-11-14T14:00:54Z

F-056 ACTIVE conf 0.98 (high) / T1070.004

SDELETE.EXE executed 7 times across two paths during the attacker's session (13:42–13:47 UTC 2020-11-14). First 2 runs from Users\Fredr\Downloads\SDelete\SDELETE.EXE (Prefetch hash -2BD91720) at 13:42:30 and 13:42:38 UTC; subsequent 5 runs with SDelete copied to...

executable: SDELETE.EXE file_path: Users\Fredr\Downloads\SDelete\SDELETE.EXE file_path: Windows\System32\SDELETE.EXE
file_path: Users\Fredr\OneDrive\Desktop\EARTHFORCE SA-26 THUNDERBOLT STAR FURY.DOCX
file_path: Users\Fredr\OneDrive\ADAMANTIUM-BACKGROUND.DOCX
file_path: Users\Fredr\OneDrive\SUCCESS-TEST-PLAN-VIBRANIUM-ALLOY-RESULTS.DOCX

F-112 ACTIVE conf 0.98 (high) / T1083

WordWheelQuery registry key (NTUSER.DAT, LastWriteTimestamp 2020-11-14 14:04:07 UTC) records 11 search terms typed into Windows Explorer's search bar, revealing the attacker's systematic intelligence-collection methodology in reverse MRU order: MRU:10 starfury, MRU:9 kitt,...

timestamp: 2020-11-14T14:04:07Z si_modified: 2020-11-14T14:04:07Z

F-100 ACTIVE conf 0.98 (high) / T1567.002

LNK file SRL-EMAIL-EXPORT.lnk (SourceCreated 2020-11-14 14:01:35 UTC) definitively proves the Outlook PST export was placed on Google Drive File Stream as G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst (VolumeLabel: Google Drive File Stream,...

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst
file_path: C:\Users\Fredr\OneDrive\Documents\Outlook Files\backup.pst timestamp: 2020-11-14T14:01:35Z
timestamp: 2020-11-14T14:03:52Z

F-048 ACTIVE conf 0.98 (high) / T1021.001

INBOUND RDP takeover from attacker host 'cobra' (52.249.198.56, Azure IP): At 2020-11-14 03:42:44 UTC (22:42 EST Nov 13) the LOCAL console session for SRL-FORGE\Fredr was disconnected (EID 4779, Address: LOCAL) and within 5 seconds was reconnected via RDP from hostname...

owner_process: 52.249.198.56 process_path: cobra timestamp: 2020-11-14T03:42:49

F-123 ACTIVE conf 0.97 (high) / T1070.004

compare_disk_and_memory (E-204) confirms two execution_evidence_deleted_binary HIGH discrepancies for SDELETE.EXE (F-056/F-115) and sdelete64.exe (F-059/F-118); Prefetch/Amcache proves both binaries executed; Recycle Bin and PowerShell history prove both were subsequently...

executable: SDELETE.EXE executable: sdelete64.exe file_path: C:\Users\Fredr\Downloads\SDelete\sdelete.exe
timestamp: 2020-11-14T13:41:24Z

F-106 ACTIVE conf 0.97 (high) / T1213.002

Jump Lists confirm the attacker accessed Maria Hill's (mhill) private SharePoint personal documents through Fred's authenticated Office 365 session, opening 5 files from mhill's SharePoint site: (1)...

file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/KITT-CompetitiveAnalysisDocs/The%20Future%20of%20KITT.pptx
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/WorkingFiles/German-KITT-Specs.docx
timestamp: 2020-11-14T03:48:53Z timestamp: 2020-11-14T04:23:05Z

F-104 ACTIVE conf 0.97 (high) / T1021.001

Jump List for Remote Desktop Connection (AppId: 1bc392b8e104a00e) proves mstsc.exe was used to RDP to "base-rd-08.shieldbase.lan" on 2020-11-14, LastModified 05:05:38 UTC with InteractionCount=4 (connected 4 times). The DestList Arguments field contains...

executable: C:\Windows\System32\mstsc.exe owner_process: mstsc.exe timestamp: 2020-11-14T05:05:38Z
path: base-rd-08.shieldbase.lan

F-101 ACTIVE conf 0.97 (high) / T1052.001

CRIMSON2 (F:\) USB drive was used as primary physical exfiltration medium. LNK files created during attack window 2020-11-14 03:46-04:30 UTC confirm the attacker brought the CRIMSON2 drive (VolumeLabel: CRIMSON2) and structured it with pre-planned folders: F:\Files from SRL...

file_path: F:\Files from SRL system file_path: F:\Files of interest\Megaforce file_path: F:\Key Data\Blue Thunder
file_path: F:\Recovered Documents\Wolves_Lair_Tech_Specs.pptx timestamp: 2020-11-14T03:46:00Z timestamp: 2020-11-14T04:30:00Z

F-049 ACTIVE conf 0.97 (high) / T1021.001

Lateral movement from Fred's laptop to shieldbase infrastructure: At 2020-11-14 05:00:44 UTC (00:00 EST) attacker used RDP Client to connect to base-rd-08.shieldbase.lan (172.16.6.18, private IP). TerminalServices-RDPClient EID 1024 (Dest: base-rd-08.shieldbase.lan) and EID...

owner_process: base-rd-08.shieldbase.lan process_path: 172.16.6.18 value_data: SHIELDBASE\frocba
timestamp: 2020-11-14T05:00:44

F-102 ACTIVE conf 0.96 (high) / T1005

LNK file analysis confirms the breadth of SRL intellectual property accessed and exfiltrated during the 2020-11-14 attack. Under fredr's profile, attacker accessed (ordered by time UTC): ADAMANTIUM-Background.docx (03:43, C:\Users\fredr\Documents\SRL);...

file_path: C:\Users\fredr\Documents\SRL\ADAMANTIUM-Background.docx
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Research to Weaponize the Ion Thruster.docx
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air\Wolf Air Financials.xlsx
file_path: C:\Users\fredr\OneDrive\Documents\Vibrainium-SRL.docx timestamp: 2020-11-14T03:43:00Z
timestamp: 2020-11-14T04:36:51Z

F-127 ACTIVE conf 0.95 (high) / T1490

analyze_vss (E-210) found 0 Volume Shadow Copies on the rocba-cdrive.e01 image. VSS is absent, meaning pre-clearing Security.evtx recovery is not possible. The absence may reflect: (1) VSS was disabled on this Windows 10 device (common on devices with small SSD storage), (2)...

timestamp: 2020-11-14T00:34:00 executable: vshadowinfo confirmed 0 shadow stores

F-059 ACTIVE conf 0.95 (high) / T1136.001

Amcache confirms the srl-h account downloaded SDelete (sdelete64.exe, SHA-1: 7bcd946326b67f806b3db4595ede9fbdf29d0c36) to c:\users\srl-h\downloads\ on 2020-10-21 03:44:22 UTC — 24 days before the break-in. The identical SHA-1 appears for...

executable: sdelete64.exe file_path: c:\users\srl-h\downloads\sdelete64.exe
file_path: c:\users\fredr\downloads\sdelete\sdelete.exe timestamp: 2020-10-21T03:44:22Z timestamp: 2020-11-14T13:42:33Z

F-122 ACTIVE conf 0.95 (high) / T1070.001

Chainsaw Sigma scan (2,284 rules, E-201) against Security.evtx, System.evtx, PowerShell.evtx, and Application.evtx produced 0 medium/high/critical detections. All 7 hits are "User Logoff Event" (T1531, informational) timestamped 2020-11-16 02:29:36-42 UTC (forensic team...

timestamp: 2020-11-14T00:34:00 executable: Security.evtx chunk parse failures chunks 254-289

F-096 ACTIVE conf 0.95 (high) / T1567.002

ShellBags confirm Google Drive File Stream was mounted as G:\ and the attacker actively browsed the following G: drive folders during the attack: G:\My Drive\STARK-RESEARCH-LABS FOLDER\ (03:57:03 UTC), G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air (04:27:21...

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air

file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER

timestamp: 2020-11-14T14:00:50Z

timestamp: 2020-11-14T03:57:03Z

F-052 ACTIVE conf 0.95 (high) / T1114.001

Attacker created backup.pst at 2020-11-14 13:39:21 UTC in Users\fredr\OneDrive\Documents\Outlook Files, wrote email data to it until 14:02 UTC, then deleted it to the Recycle Bin (\$RDNBREY.pst) at 14:07:33 UTC. USN confirms FileCreate → DataOverwrite sequence followed by...

file_path: Users\fredr\OneDrive\Documents\Outlook Files\backup.pst

file_path: \$Recycle.Bin\RDNBREY.pst

file_path: \$Recycle.Bin\IDNBREY.pst

timestamp: 2020-11-14T13:39:21Z

timestamp: 2020-11-14T14:07:33Z

F-047 ACTIVE conf 0.95 (high) / T1110.003

NTLM brute force campaign: 58,713 EID 4776 (NTLM validation) and 58,706 EID 4625 (failed logon) events in Security.evtx spanning 2020-11-13 to 2020-11-14 UTC. Targets include ADMINISTRATOR (0xC000006A = correct username, wrong password — account confirmed to exist), and...

executable: ADMINISTRATOR

timestamp: 2020-11-13T00:00:01

F-118 ACTIVE conf 0.92 (high) / T1070.004

srl-h's PowerShell history records sequential execution of sdelete64.exe -nobanner -z -c D: and sdelete64.exe -nobanner -z -c C:, zeroing all free space on both C: and D: drives to frustrate file-recovery forensics. No fredr PowerShell history exists, confirming the break-in...

executable: sdelete64.exe

process_path: C:\Users\srl-h\Downloads\sdelete64.exe

timestamp: 2020-10-21T03:46:19

F-117 ACTIVE conf 0.92 (high) / T1021.001

srl-h PowerShell history (mtime 2020-10-21T03:46:19 UTC) records 6 commands: initial attempts to run sdelete64.exe via UNC tsclient path \\tsclient\C\Users\rsydow\Documents\SRL Admin\Workstation-Prep\sdelete64.exe targeting D: drive, followed by cd to srl-h Downloads and...

path: \\tsclient\C\Users\rsydow\Documents\SRL Admin\Workstation-Prep\sdelete64.exe

executable: sdelete64.exe

process_path: C:\Users\srl-h\Downloads\sdelete64.exe

timestamp: 2020-10-21T03:46:19

Corroboration & Suggested Next Steps (advisory)

Advisory only - does not change finding_status or confidence. Distinct from ATT&CK *Suggested Next Tools* (tactic blind-spots) and *Active Leads* (detector pivots); this is forensic-knowledge corroboration - what would raise each finding to the next evidentiary tier.

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
F-001	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-002	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-003	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-004	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-005	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-006	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-007	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-008	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-009	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-010	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-011	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-012	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-013	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-014	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-015	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-016	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-017	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-018	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-019	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-020	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-021	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-022	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-023	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-024	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-025	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-026	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-027	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-028	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-029	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-030	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-031	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-032	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-033	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-034	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-035	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-036	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-037	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-038	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-039	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline (encoded/hidden args)	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone -> observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-040	volatility_memory	observation	unknown	memory netscan (PID->C2 socket), memory cmdline	memory.scan_processes	advisory only; does not reflect finding_status or promotion. memory_process alone ->

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
				(encoded/hidden args)		observation (no corroborating source found). To strengthen, corroborate with: memory netscan (PID->C2 socket); memory cmdline (encoded/hidden args).
F-041	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-042	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-043	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-044	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-045	mft	observation	unknown	mft (\$SI/\$FN)	disk.extract_mft_timeline	advisory only; does not reflect finding_status or promotion. mft alone -> observation (no corroborating source found). To strengthen, corroborate with: mft (\$SI/\$FN).
F-046	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen,

FINDING	ARTIFACT	ADVISORY TIER	TS ALIGNED	GAP	SUGGESTED TOOLS	RATIONALE
						corroborate with: EVTX 4688 process creation; Prefetch execution.
F-047	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-048	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-049	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.
F-050	event_logs_security	observation	unknown	EVTX 4688 process creation, Prefetch execution	disk.summarize_evtx, disk.extract_prefetch	advisory only; does not reflect finding_status or promotion. evtx_process_creation alone -> observation (no corroborating source found). To strengthen, corroborate with: EVTX 4688 process creation; Prefetch execution.

Indicators of Compromise

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST\SRL-EMAIL-EXPORT.pst	path/file	F-125	CONFIRMED	T1114.001
file_path: C:\Users\fredr\OneDrive\Documents\Outlook Files\backup.pst	path/file	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:01:34	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:03:52	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T14:04:07	indicator	F-125	CONFIRMED	T1114.001
timestamp: 2020-11-14T03:42:49	indicator	F-124	CONFIRMED	T1005
timestamp: 2020-11-14T03:46:33	indicator	F-124	CONFIRMED	T1005
executable: CRIMSON2 USB volume mounted	executable	F-124	CONFIRMED	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\SRL-Projects	path/file	F-124	CONFIRMED	T1005
file_path: C:\Users\fredr\Downloads\SDelete\sdelete.exe	path/file	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T14:07:32Z	indicator	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T13:41:24Z	indicator	F-115	ACTIVE	T1070.004
timestamp: 2020-11-14T04:49:53Z	indicator	F-115	ACTIVE	T1070.004
file_path: C:\Users\fredr\OneDrive\Desktop\BitLocker Recovery Key 26F77152-999C-45E8-8BD4-C83FAC7BB72D.TXT	path/file	F-114	ACTIVE	T1552.001
timestamp: 2020-11-14T12:48:13Z	indicator	F-114	ACTIVE	T1552.001
file_path: F:\Files from SRL system\Quantum Particles Affected by Other Dimensions.pdf	path/file	F-109	ACTIVE	T1213.002
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/Quantum%20Particles%20Affected%20by%20Other%20Dimensions.pdf	indicator	F-109	ACTIVE	T1213.002
timestamp: 2020-11-14T04:13:22Z	indicator	F-109	ACTIVE	T1213.002
executable: C:\Users\fredr\Downloads\SDelete.zip	executable	F-108	ACTIVE	T1070.004
file_path: C:\Users\fredr\Downloads\SDelete.zip	path/file	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T13:37:51Z	indicator	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T13:37:37Z	indicator	F-108	ACTIVE	T1070.004
timestamp: 2020-11-14T14:01:34Z	indicator	F-105	ACTIVE	T1114.001
timestamp: 2020-11-14T14:03:52Z	indicator	F-105	ACTIVE	T1114.001
timestamp: 2020-11-14T14:00:54Z	indicator	F-105	ACTIVE	T1114.001
executable: SDELETE.EXE	executable	F-056	ACTIVE	T1070.004
file_path: Users\fredr\Downloads\SDelete\SDELETE.EXE	path/file	F-056	ACTIVE	T1070.004

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
file_path: Windows\System32\SDELETE.EXE	path/file	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\Desktop\EARTHFORCE SA-26 THUNDERBOLT STAR FURY.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\ADAMANTIUM-BACKGROUND.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\SUCCESS-TEST-PLAN-VIBRANIUM-ALLOY-RESULTS.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\THE SHIELD BACKGROUND AND ONGOING RESEARCH.DOCX	account	F-056	ACTIVE	T1070.004
file_path: Users\fredr\OneDrive\NOKIA STRATEGY.DOCX	account	F-056	ACTIVE	T1070.004
timestamp: 2020-11-14T13:42:30Z	indicator	F-056	ACTIVE	T1070.004
timestamp: 2020-11-14T14:04:07Z	indicator	F-112	ACTIVE	T1083
si_modified: 2020-11-14T14:04:07Z	indicator	F-112	ACTIVE	T1083
timestamp: 2020-11-14T14:01:35Z	indicator	F-100	ACTIVE	T1567.002
owner_process: 52.249.198.56	process	F-048	ACTIVE	T1021.001
process_path: cobra	indicator	F-048	ACTIVE	T1021.001
executable: sdelete64.exe	executable	F-123	ACTIVE	T1070.004
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/KITT-CompetitiveAnalysisDocs/The%20Future%20of%20KITT.pptx	indicator	F-106	ACTIVE	T1213.002
file_path: https://starkresearchlabs-my.sharepoint.com/personal/mhill_stark-research-labs_com/Documents/WorkingFiles/German-KITT-Specs.docx	indicator	F-106	ACTIVE	T1213.002
timestamp: 2020-11-14T03:48:53Z	indicator	F-106	ACTIVE	T1213.002
timestamp: 2020-11-14T04:23:05Z	indicator	F-106	ACTIVE	T1213.002
executable: C:\Windows\System32\mstsc.exe	executable	F-104	ACTIVE	T1021.001
owner_process: mstsc.exe	process	F-104	ACTIVE	T1021.001
timestamp: 2020-11-14T05:05:38Z	indicator	F-104	ACTIVE	T1021.001
path: base-rd-08.shieldbase.lan	indicator	F-104	ACTIVE	T1021.001
file_path: F:\Files from SRL system	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Files of interest\Megaforce	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Key Data\Blue Thunder	path/file	F-101	ACTIVE	T1052.001
file_path: F:\Recovered Documents\Wolves_Lair_Tech_Specs.pptx	path/file	F-101	ACTIVE	T1052.001
timestamp: 2020-11-14T03:46:00Z	indicator	F-101	ACTIVE	T1052.001
timestamp: 2020-11-14T04:30:00Z	indicator	F-101	ACTIVE	T1052.001
owner_process: base-rd-08.shieldbase.lan	process	F-049	ACTIVE	T1021.001

INDICATOR	TYPE	SOURCE FINDING	STATUS	TECHNIQUE
process_path: 172.16.6.18	ip	F-049	ACTIVE	T1021.001
value_data: SHIELDBASE\frocba	account	F-049	ACTIVE	T1021.001
timestamp: 2020-11-14T05:00:44	indicator	F-049	ACTIVE	T1021.001
file_path: C:\Users\fredr\Documents\SRL\ADAMANTIUM-Background.docx	path/file	F-102	ACTIVE	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Research to Weaponize the Ion Thruster.docx	path/file	F-102	ACTIVE	T1005
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air\Wolf Air Financials.xlsx	path/file	F-102	ACTIVE	T1005
file_path: C:\Users\fredr\OneDrive\Documents\Vibrainium-SRL.docx	path/file	F-102	ACTIVE	T1005
timestamp: 2020-11-14T03:43:00Z	indicator	F-102	ACTIVE	T1005
timestamp: 2020-11-14T04:36:51Z	indicator	F-102	ACTIVE	T1005
timestamp: 2020-11-14T00:34:00	indicator	F-127	ACTIVE	T1490
executable: vshadowinfo confirmed 0 shadow stores	executable	F-127	ACTIVE	T1490
file_path: c:\users\srl-h\downloads\sdelete64.exe	path/file	F-059	ACTIVE	T1136.001
file_path: c:\users\fredr\downloads\sdelete\sdelete.exe	path/file	F-059	ACTIVE	T1136.001
timestamp: 2020-10-21T03:44:22Z	indicator	F-059	ACTIVE	T1136.001
timestamp: 2020-11-14T13:42:33Z	indicator	F-059	ACTIVE	T1136.001
executable: Security.evtx chunk parse failures chunks 254-289	executable	F-122	ACTIVE	T1070.001
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Exported-PST	path/file	F-096	ACTIVE	T1567.002
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER\Airwolf-SRL\Wolf Air	path/file	F-096	ACTIVE	T1567.002
file_path: G:\My Drive\STARK-RESEARCH-LABS FOLDER	path/file	F-096	ACTIVE	T1567.002
timestamp: 2020-11-14T14:00:50Z	indicator	F-096	ACTIVE	T1567.002
timestamp: 2020-11-14T03:57:03Z	indicator	F-096	ACTIVE	T1567.002
file_path: Users\fredr\OneDrive\Documents\Outlook Files\backup.pst	account	F-052	ACTIVE	T1114.001
file_path: \$Recycle.Bin\RDNBREY.pst	account	F-052	ACTIVE	T1114.001
file_path: \$Recycle.Bin\IDNBREY.pst	account	F-052	ACTIVE	T1114.001
timestamp: 2020-11-14T13:39:21Z	indicator	F-052	ACTIVE	T1114.001

Anti-Forensics Warnings

- None recorded.

Data Gaps

- **warning:** Terminal absence: 0 shadow copies on rocba-cdrive.e01 — pre-clearing Security.evtx events (before 2020-11-14 00:34 UTC) are permanently irrecoverable; attacker log clearing, initial access, and credential-theft events cannot be recovered
- **warning:** Cannot determine whether VSS absence is from configuration, storage limits, or attacker deletion — EID 4688 for vssadmin/wmic unavailable due to cleared Security.evtx

TECHNICAL APPENDIX

Sigma Anomaly Summary

#	Severity	Detector	Description	ATT&CK
1	MEDIUM	motw_presence	Mark-of-the-Web Zone.Identifier evidence is present.	T1204

Recorded Hunting Hypotheses

Each hypothesis formed during the hunt and its verdict after testing against the evidence: proven (CONFIRMED), disproven (REFUTED), or inconclusive (SUSPENDED). "Linked finding IDs" are the F-NNN that proved, refuted, or materially informed the verdict.

HYPOTHESIS ID	ATTACK CLASS	VERDICT	LINKED FINDING IDS	MITRE
01KTJCM8EVCN629KMMCX2V7C81	pre_planned_physical_intrusion_staged_account (rank 1)	CONFIRMED - proven	F-046, F-047, F-057, F-059, F-117, F-118, F-101, F-102, F-124	-
01KTJCM8EVAZZC9GZ1Y16S5BNV	multi_channel_exfiltration_usb_cloud_lateral_movement (rank 2)	CONFIRMED - proven	F-048, F-049, F-050, F-100, F-101, F-102, F-104, F-105, F-106, F-108, F-109, F-110, F-112, F-115, F-124, F-125	-
01KTJCM8EV2X43RSP45X7XXDHQ	bitlocker_recovery_key_exfiltration (rank 3)	SUSPENDED - inconclusive	F-112, F-114	-
01KTJCM8EVYAD4MTFA8X3CZARX	icloud_opportunistic_cloud_sync_exfiltration (rank 4)	SUSPENDED - inconclusive	F-038	-
01KTJCM8EV2D5MX77J654SNF78	anti_forensics_evidence_destruction_multi_method (rank 5)	SUSPENDED - inconclusive	F-053, F-054, F-056, F-059, F-108, F-114, F-115, F-118, F-122, F-123	-

Analysis Lanes

LANE	STATUS	REQUIRED	AGENTS	EXECUTIONS	FINDINGS	SUMMARY
evidence_access	COMPLETE	yes	main-agent	0	0	load_memory: Memory dump ready at /evidence/memory/Rocba-Memory.raw
memory	COMPLETE	yes	main-agent	14	40	2186 processes enumerated; no DKOM hiding, no confirmed code injection (0 MZ headers); full iCloud suite active (APSDaemon, iCloudDrive, iCloudServices, iCloudIE) all loading...
disk_execution_persistence	COMPLETE	yes	main-agent	52	60	PowerShell history proves RDP tsclient access from rsydow's machine and sdelete64 free-space wiping; scheduled tasks show no attacker persistence; all mandatory file-access...
event_auth	COMPLETE_WITH_GAPS	yes	main-agent	21	6	EVTX analysis confirms full attacker lifecycle: 58K+ NTLM brute force (T1110.003), RDP session takeover from 'cobra' (52.249.198.56) at 22:42 EST (T1021.001), lateral movement...
anti_forensics_recovery	COMPLETE_WITH_GAPS	yes	main-agent	1	5	analyze_vss found 0 shadow copies (E-210) — VSS recovery of pre-clearing Security.evtx is impossible; F-127 documents the terminal evidence gap as a permanent irrecoverable...
timeline_correlation	COMPLETE_WITH_GAPS	yes	main-agent	25	14	Sigma scan (2,284 rules, E-201) yielded 0 medium/high/critical hits; Security.evtx had 12 corrupt chunks confirming log clearing — the negative Sigma result corroborates...
synthesis_corroboration	COMPLETE	yes	main-agent	2	3	compare_disk_and_memory (15 discrepancies, all explained) and find_temporal_clusters (5 clusters) promoted two CONFIRMED findings: active SRL IP collection at 03:42-03:46 UTC...

Orchestration Warnings

- None recorded.

Open Questions

- No open questions recorded.

Status Flags

```
{
  "open_leads": true,
  "anti_forensics_warning": false,
  "unresolved_discrepancy": true,
  "specialist_lanes_inferred": false,
  "hypotheses_unresolved": 0
}
```

Findings Status Breakdown

ACTIVE

125

CONFIRMED

2